

$$f(x) = \frac{1}{x}$$

$$g(x) = x^2$$

$$h(x) = \frac{x^2}{8}$$

IRUDIKAPENA

 $f(x) = \frac{1}{x}$ Aldearautik funtzioa $g(x) = x^2$ Parabola $\in (0, \infty)$ $h(x) = x^2/8$ Parabola $\in (0, \infty)$

x	1	0.5	2
y	1	2	0.5

x	1	2
y	1	4

x	0	1	2
y	0	$\frac{1}{8}$	$\frac{1}{2}$

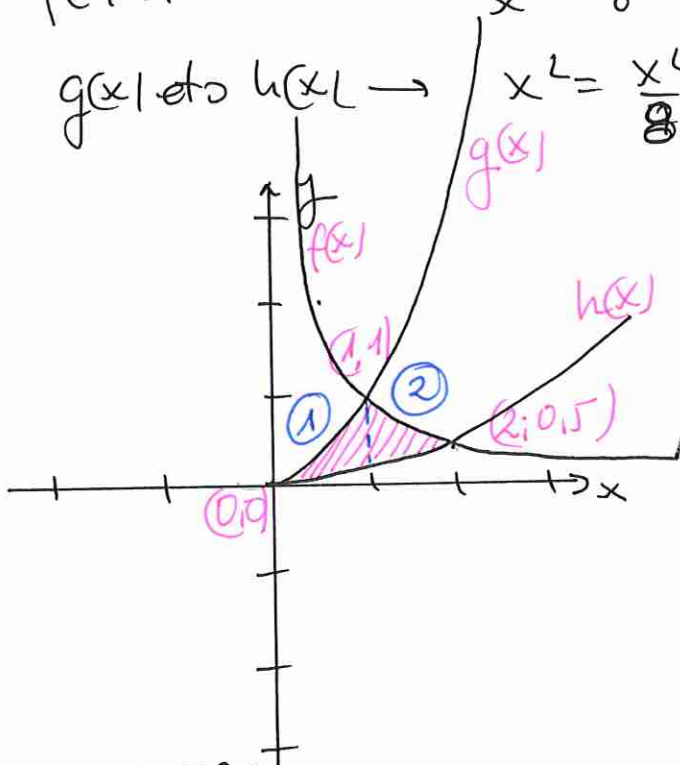
1.) EBAKI PUNTUAK.

$$f(x) \text{ eta } g(x) \rightarrow \frac{1}{x} = x^2 \quad x^3 = 1$$

$$x = 1 \quad y = 1 \quad \boxed{(1, 1)}$$

$$f(x) \text{ eta } h(x) \rightarrow \frac{1}{x} = \frac{x^2}{8} \rightarrow x^3 = 8 \quad \boxed{x = 2 \quad y = 0.5}$$

$$g(x) \text{ eta } h(x) \rightarrow x^2 = \frac{x^2}{8} \rightarrow \frac{7}{8}x^2 = 0 \quad \boxed{x = 0 \quad y = 0}$$

2.) Kalkula funtzioak

$$\textcircled{1} \text{ esparrua } g(x) - h(x) =$$

$$= x^2 - \frac{x^2}{8} = \frac{7}{8}x^2$$

$$\textcircled{2} \text{ esparrua } f(x) - h(x) =$$

$$= \frac{1}{x} - \frac{x^2}{8}$$

3. ALAKERA

$$A = \int_0^1 \frac{7}{8}x^2 dx + \int_1^2 \left(\frac{1}{x} - \frac{x^2}{8} \right) dx =$$

$$= \left[\frac{7}{8} \frac{x^3}{3} \right]_0^1 + \left[\ln x - \frac{x^3}{24} \right]_1^2 = \frac{7}{24} + \left(\ln 2 - \frac{8}{24} \right) - \left(\ln 1 - \frac{1}{24} \right) =$$

$$= \frac{7}{24} + \ln 2 - \frac{8}{24} + \frac{1}{24} = \boxed{\ln 2 + \frac{1}{24}}$$

2021 @ DHIKOA AU

$$y = 4x - x^2$$

$$y = x^2 - 6$$

1) EBAKI PUNTUAK

$$4x - x^2 = x^2 - 6$$

$$0 = 2x^2 - 4x - 6$$

$$x = \frac{4 \pm \sqrt{4^2 - 4 \cdot 2 \cdot (-6)}}{2 \cdot 2} = \frac{4 \pm 8}{4}$$

$$x_1 = 3 \quad y_1 = 3$$

$$x_2 = -1 \quad y_2 = -5$$

$$\boxed{\begin{matrix} (3, 3) \\ (-1, -5) \end{matrix}}$$

Irudi kopuru

$$f(x) = 4x - x^2$$

$$E(+2, 4)$$

Ox ardatzotzat:

$$0 = 4x - x^2$$

$$0 = x(4 - x)$$

$$x_1 = 0 \quad (0, 0)$$

$$x_2 = 4 \quad (4, 0)$$

$$g(x) = x^2 - 6$$

$$U \quad E(0, -6)$$

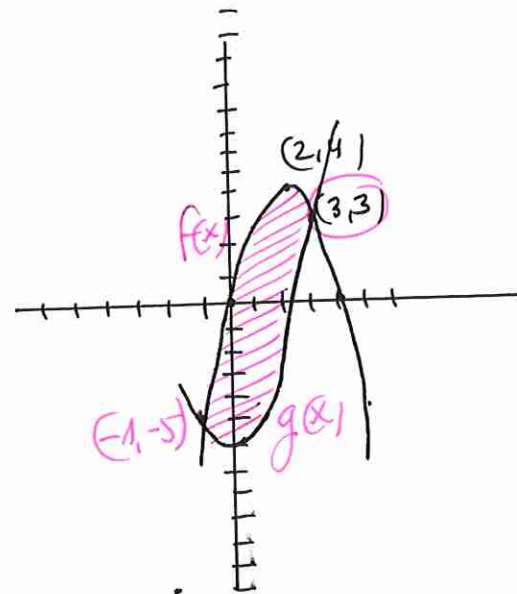
Ox ardatzotzat:

$$0 = x^2 - 6$$

$$x = \pm \sqrt{6}$$

$$(\sqrt{6}, 0)$$

$$(-\sqrt{6}, 0)$$



2.) Kurketo funtzioa

$$h(x) = f(x) - g(x) = (4x - x^2) - (x^2 - 6)$$

$$h(x) = -2x^2 + 4x + 6$$

3.) Azaleria

Barrowen erregelak

$$\int_{-1}^3 h(x) dx = H(3) - H(-1)$$

$$A = \int_{-1}^3 (-2x^2 + 4x + 6) dx = \left[-\frac{2x^3}{3} + 2x^2 + 6x \right]_{-1}^3 =$$

$$= \left(-\frac{2 \cdot 3^3}{3} + 2 \cdot 3^2 + 6 \cdot 3 \right) - \left(-\frac{2 \cdot (-1)^3}{3} + 2(-1)^2 + 6 \cdot (-1) \right) =$$

$$= (-18 + 18 + 18) - \left(\frac{2}{3} + 2 - 6 \right) = 18 - \frac{2}{3} + 4 =$$

$$= 22 - \frac{2}{3} = \boxed{\frac{64}{3} \text{ u}^2}$$

2020 OHIKOA A4

$$f(x) = x^2 - 2x + 1$$

$$g(x) = -x^2 + 5$$

1.) EBAKI-PUNTUAK

$$x^2 - 2x + 1 = -x^2 + 5$$

$$2x^2 - 2x - 4 = 0$$

$$x^2 - x - 2 = 0$$

$$x = \frac{1 \pm \sqrt{1 - 4(-2)}}{2}$$

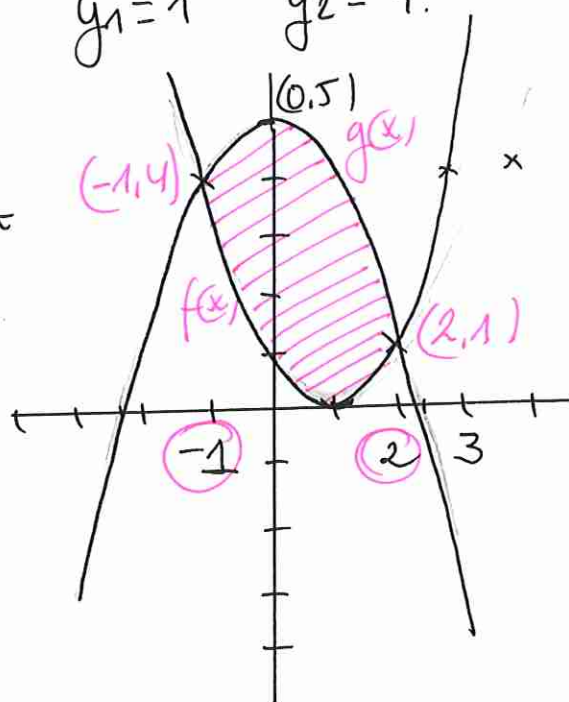
$$x = \frac{1 \pm 3}{2}$$

$$x_1 = 2$$

$$y_1 = 1$$

$$x_2 = -1$$

$$y_2 = 4$$



Irudi kopuru

$$f(x) = x^2 - 2x + 1$$

$$E(1, 0)$$

ox ardatziko (1, 0)

$$g(x) = -x^2 + 5$$

$$E(0, 5)$$

Ox ardatziko

$$x_1 = \sqrt{5}$$

$$x_2 = -\sqrt{5}$$

2.) Kenketako funtzioa

$$h(x) = g(x) - f(x)$$

$$h(x) = (-x^2 + 5) - (x^2 - 2x + 1)$$

$$h(x) = -2x^2 + 2x + 4$$

barne-behako

3.) Azalera

$$A = \int_{-1}^2 h(x) dx = H(2) - H(-1)$$

BARROWEN
ERREJENA

$$A = \int_{-1}^2 (-2x^2 + 2x + 4) dx = \left[-\frac{2x^3}{3} + x^2 + 4x \right]_{-1}^2 =$$

$$= \left(-\frac{2 \cdot 2^3}{3} + 2^2 + 4 \cdot 2 \right) - \left(-\frac{2(-1)^3}{3} + (-1)^2 + 4(-1) \right) =$$

$$= -\frac{16}{3} + 4 + 8 - \frac{2}{3} - 1 + 4 = 15 - \frac{18}{3} = 15 - 6 = \boxed{9u^2}$$

$$y = 3 - x^2$$

$$y = 2x$$

1. FAK-PUNTNAK

$$3 - x^2 = 2x$$

$$0 = x^2 + 2x - 3$$

$$x = \frac{-2 \pm \sqrt{2^2 - 4 \cdot (-3)}}{2} = \frac{-2 \pm 4}{2}$$

$$x_1 = -3 \quad y_1 = -6$$

$$x_2 = 1 \quad y_2 = 2$$

$(-3, -6)$
 $(1, 2)$

Indikopero

$$f(x) = 3 - x^2$$

$$E(0, 3) \quad \uparrow$$

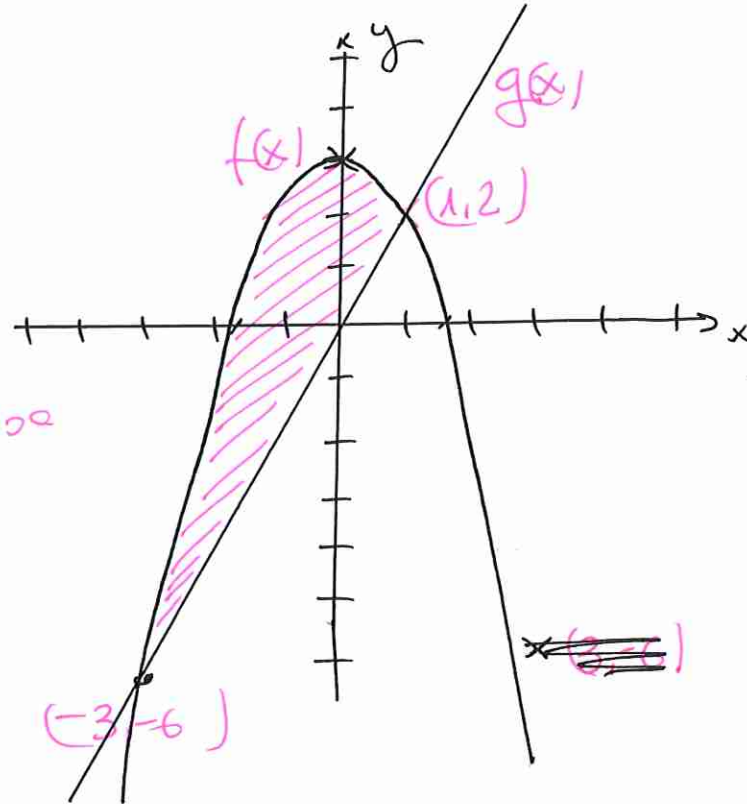
Oxardtopot

$$0 = 3 - x^2$$

$$x = \pm \sqrt{3}$$

$$g(x) = 2x$$

x	y
0	0
1	2



2.) Kunkto funktsioa

Goko funktsioa - Beluk funktsioa

$$h(x) = f(x) - g(x) =$$

$$h(x) = (3 - x^2) - 2x \Rightarrow$$

$$h(x) = -x^2 - 2x + 3$$

3.) Azalena

Barrowen erregel

$$\int_{-3}^1 (-x^2 - 2x + 3) dx = H(1) - H(-3)$$

$$\int_{-3}^1 (-x^2 - 2x + 3) dx = \left[-\frac{x^3}{3} - x^2 + 3x \right]_{-3}^1 =$$

$$= \left(-\frac{1^3}{3} - 1^2 + 3 \cdot 1 \right) - \left(-\frac{(-3)^3}{3} - (-3)^2 + 3(-3) \right) =$$

$$= -\frac{1}{3} - 1 + 3 - \frac{27}{3} + 9 + 9 = -\frac{28}{3} + 20 =$$

$\frac{32}{3}$